



Project Completion Report

(Dead-man-switch) Permission based Wallet, shipped as Epora

Project Name	(Dead-man-switch) Permission based Wallet, shipped as Epora
Project URL	projectcatalyst.io , Fund 11, Cardano Use Cases: Concept
Project Number	1100002
Project Manager	Sandro Schaier
Project Start Date	11 March 2024
Project Completion Date	26 September 2026
Plan	7 months, with deliveries from April to October 2024
Budget	100,000 ADA in five milestones
Completion video	Watch the close-out video on YouTube

1 Deliverables

Epora is a permission-based smart wallet on Cardano. It lets users plan recovery after inactivity, share control, and give other people limited spending rights. A normal wallet holds the signing keys. The smart wallet holds funds and checks whether each action meets its configured rules.

The project delivered the contract system, reference application, public API, whitepaper, and developer documentation. The application runs on [mainnet](#) as an unaudited beta, with a [Preprod deployment](#) for testing. The project's own code and documents are public under the [MIT license](#).

What was built

Component	Delivered functionality and evidence
Permission contracts	Aiken validators enforce owner and weighted multisignature actions, per-day allowances, proof-of-life renewal, weighted beneficiary recovery, and streaming payments. The system also supports consolidation, stake control, reward withdrawal, certificates, and governance votes. See the transition map .
Reference application	A web interface to create and fund smart wallets, configure permissions, review transactions, and sign through a connected wallet. Co-signers can review and complete shared approval requests. See the application guide and live beta .
Public API	A versioned HTTP API gives third-party developers access to indexed wallet state and unsigned transaction builders. Signing remains with the user's wallet. The developer guide , interactive reference , and OpenAPI specification document the interface.
Design and security documentation	The whitepaper explains the architecture, permission model, use cases, and security assumptions. The security evidence map connects threats to executable tests.

Feedback and wallet outreach

The [feedback register](#) links reports to fixes or a deferral. Its 22 September snapshot contains 49 distinct issues: 46 categorized as fixed, one deferred, and two non-bugs. External GitHub and Discord reports led to wallet-connection and interface improvements. The live VESPR retest is complete.

Eternl, Begin Wallet, and other wallet developers were contacted about native integration and a possible smart-wallet CIP. Eternl replied by direct message. The [outreach record](#) includes the public Discord message and a screenshot. Communication is complete; integration discussions continue.

Project demonstration. [Watch the Epora Wallet Catalyst-Closeout video](#). The [Milestone 5 submission](#) maps each final output to its acceptance criteria and evidence.

2 Usage

Who the wallet is for

Owners can configure a recovery plan if they become inactive. Families can name beneficiaries and assign recovery shares. Shared treasuries can require co-signatures or give users daily allowances. Developers can use the API to build wallet integrations.

A user connects a signing wallet, creates and funds a smart wallet, then sets its permissions. Each action follows a review and signature flow. A co-signed action collects the required approval weight. When the proof-of-life period expires, eligible beneficiaries can use the configured recovery path. These are intended uses; the activity snapshot below measures the deployed prototype.

Mainnet activity: 26 September 2026, 01:30 UTC

The current deployed policy identifies two active smart wallets. The two wallets held four unspent wallet outputs. These figures describe prototype activity under the current policy, not independent users or adoption across earlier contract versions.

Wallet	Wallet ADA	State deposit ADA	Total ADA
Smart wallet	10.000000	1.586080	11.586080
Smart wallet Demo	17.000000	2.202410	19.202410
Total	27.000000	3.788490	30.788490

Neither wallet payment balance contains other native tokens. Each separate state output holds its identifying state-thread token. The shared reference output holds another 63.270800 ADA. These deposits are separate from user wallet balances and are not fees or revenue.

Observed actions and measured fees

Seven wallet transactions and one reference deployment are confirmed on mainnet. The table lists the actual ledger fees and transaction sizes. Each link opens the corresponding transaction.

Confirmed action	Fee ADA	Bytes	Transaction
Reference deployment	0.809617	14868	fb1186638076...
Wallet creation + funding	0.447150	838	71ea33e0e820...
Wallet additional funding	0.170825	350	d6e2e824daef...
Wallet operator spend	0.895889	10117	8d21e3b7f954...
Demo creation + funding	0.447194	839	6ba0055f3cdd...
Demo additional funding	0.170825	350	9e60d91d7100...
Demo beneficiary + liveness settings	0.478150	919	4080136da4b6...
Demo spend + liveness refresh	0.912502	10256	a9502540ad8e...

The Demo spend paid 10 ADA to its recipient and returned 10 ADA to the smart wallet. An earlier 7 ADA output remained unspent. The proof-of-life deadline moved from 25 October 2026, 23:27:06.972 UTC to 23:33:36 UTC. This verifies automatic refresh during an operator spend.

Current STT policy and validator hash:

0dac00be80879dcf585cdb9d0acf6e0ecf52c417ded30d8b72d0ebf1

Evidence. The [mainnet evidence record](#) provides full hashes, addresses, the Koios query method, and measurement limits. The [JSON snapshot](#) preserves the results.

3 Impact

The project turns the proposed dead-man-switch and permission model into working contracts, an application, and an API. Users can combine recovery, shared control, and spending limits in one smart wallet. Other developers can inspect the implementation and build on the same public specification.

Project outcomes

Original objective	Result and evidence
Recover after inactivity	A configurable proof-of-life deadline and beneficiary rules. The mainnet settings transaction added a beneficiary and a 30-day period; a later operator spend refreshed the deadline. See transaction evidence .
Share control and limit spending	Weighted multisignature and per-day allowances are implemented in the contract and reference application. The whitepaper and transition map explain the authorization rules.
Deliver a usable prototype	Public Preprod and mainnet applications, a UI walkthrough, an integration API, and developer documentation. The feedback register records fixes across connection handling, transaction flows, interface behavior, and operations.
Support further development	Public source, an MIT license, a security evidence map, and wallet-developer outreach provide a basis for reuse and future integration.

Testing and measured results

The [25 September release record](#) records 750 passed, 0 failed with Aiken v1.1.23: 713 unit tests and 37 property tests. The property run used 10,000 cases per property. The rebuilt blueprint matched the committed artifact. The linked security tests cover the stated invariants and abuse cases; the beta has not had an external audit.

Mainnet transactions establish successful wallet creation, funding, settings changes, and operator spending. The Demo wallet creation cost 0.447194 ADA. Its operator spend with automatic proof-of-life refresh cost 0.912502 ADA. These are measured examples, not a fee bound for all configurations. The two-wallet inventory is not a measure of independent adoption.

Lessons from delivery

Script size affected the deployment design. The state script is 14,469 bytes. A reference script lets transactions use it without including the full script each time. The [release record](#) documents its hash and size, and the mainnet evidence records the deployed reference.

Shared control needs a usable coordination flow. A tester could not tell a co-signer to complete registration. The resulting changes added an invitation link and a per-co-signer sign-in status. The [feedback register](#) links the report to the fixes. Permissions need both enforceable rules and a clear way for people to complete the required steps.

4 Sustainability

Maintenance and operating costs

Epora continues as an open-source project. The beta is maintained, and public issues are reviewed for further fixes. The public repository and documentation allow other developers to inspect and contribute to the work.

No revenue is currently generated. Server and other operating costs are covered by received Catalyst funds. Future funding or paid services will depend on adoption and partner agreements; there is no committed recurring revenue model yet.

Next steps

Continue the beta with its unaudited-code notice and address feedback through the public issue tracker. Follow up with wallet teams and explore a smart-wallet CIP. The proposed standard could define how wallets discover smart wallets, display their rules and balances, and build permitted actions. A standard could also support shared maintenance across implementations. After community feedback on the proposed CIP, the need for an external audit will be assessed against that feedback and the CIP implementation. If an audit is needed, a separate funding round is planned to cover its cost. Existing Catalyst funds and personal funds are not intended to fund the audit.

WalletConnect signing and device testing, plus wallet-signed terms acceptance, are deferred. The beta currently uses an acknowledgement flow. The [engineering checklist](#) also retains operational follow-up: backup and reindex drills, monitoring verification, and wider fee and multi-asset testing. Those checks are separate from the delivered prototype and measured transactions in this report.

Delivery timeline and funding

The original plan covered seven months, ending in October 2024. Development extended beyond that schedule. The completion date is 26 September 2026. The project budget is 100,000 ADA; the Milestone Module showed 59,000 ADA distributed on that date.

Milestone	Delivery	Status on 26 Sep 2026
1. Planning	Social accounts, whitepaper, and development tasks	Approved 7 Jul
2. Contracts	On-chain implementation, documentation, security evidence	Approved 6 Aug
3. Interface	UI walkthrough, application, API, and code documentation	Approved 21 Sep
4. Testnet	Prototype, categorized feedback, and fixes	Submitted, in review
5. Closeout	Mainnet beta, wallet communication, report, and video	This submission

Project resources

[Public repository and documentation](#) [Mainnet beta](#) [Completion video](#)

[Mainnet evidence](#) [Feedback and fixes](#) [Wallet outreach](#)

Project 1100002. Report dated 26 September 2026. Mainnet amounts are a dated snapshot. Full transaction hashes, source links, and measurement methods accompany the report.